

L1.7b Related Rates

Recall: L1.7a Implicit Differentiation

Name _____

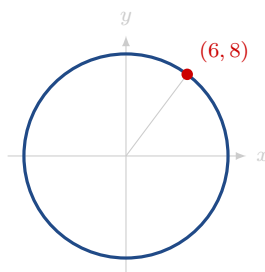
Solo – 9 minutes

1. Find y' .

(a) $x^2 + y^2 = 100$

(b) $y^2x = 4$

2. The point $(6, 8)$ sits on the circle $x^2 + y^2 = 100$. Use your answer to 1(a) to find the equation of the tangent line there.



3. A different curve has $y' = \frac{3-x}{2y+4}$.

Where is its tangent line **horizontal**? Where is it **vertical**? (Give the condition on x or on y — you are not being asked for the points.)